

A Mathematical Explanation of How Parameters Are Balanced in Zenless Zone Zero's Damage Calculations

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Introduction

ZZZ base damage: the sum of attack power, damage bonuses, critical damage, etc.



The better balance between attack power, damage bonuses, and critical damage, the higher damage



What does “good balance” mean mathematically?



The concept of majorization is useful

Objectives of This Session

1. Provide a mathematical explanation of how balance works
2. Determine the conditions under which damage increases when balancing using sub-stats, two sets of discs, and main stats 4 and 5
 - This covers ATK, Damage Bonus, and Critical Damage
 - For a detailed explanation that includes status effect damage and Sheer Force damage, please refer to the PDF

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Mathematical Background

In this section, we will focus on expressions represented as the product of several parameters.

Related Topics

Given that the height of a rectangle is x and the width is y , how can we maximize the area $S(x,y) = xy$?

At first glance, the answer seems simple: just increase the values of x and y .

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Set the sum of the side lengths, $x + y = k$, to a constant.

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Find a balance while keeping this sum constant.

Balancing

For any pair (x, y) , simply shift the values from the larger to the smaller.

For example, if $x > y$, shift by $\delta > 0$ to obtain $(x - \delta, y + \delta)$.

However, since $x - \delta > y + \delta$, the order relationship remains unchanged.

Then,

$$(x - \delta)(y + \delta) = xy + \delta(x - y - \delta) > xy + \delta^2 > xy$$

holds.

By repeating this operation multiple times, the product increases while preserving the sum.

For three variables

- To balance the set (x, y, z) , simply transfer values from the larger to the smaller of the two elements.
- **Be careful not to alter the order of magnitude before and after the transfer.**
- The more balanced the system becomes, the larger the product xyz becomes while maintaining the sum.

Example

As you move to the right, the balance is restored and the product increases.

Group	(10,1,1)	(6,5,1)	(6,4,2)	(4,4,4)
Total	10	30	48	64

- This operation is based on the T-transformation in majorization theory.
- If the goal is simply to find the maximum value, this can also be achieved using the arithmetic-geometric mean or the Lagrange multiplier method.
- However, when the condition of increasing the value is considered, majorization theory is particularly useful.

Zenless Zone Zero Damage Formula

Main Types of Damage in ZZZ

- **Basic Damage**
- Anomaly Damage
- Disorder Damage
- Sheer Force Damage

Formula for Basic Damage

Let X be ATK, Y be Damage Bonus, Z be Critical Damage, and C be a constant representing the product of these and other parameters.

Basic damage is expressed as follows:

$$D(X,Y,Z) = CXYZ.$$

In this case, we will balance the stats using the sub-stats on the Drive Disk, the two-set effect, and the main stats in slots 4 and 5.

How Parameters Are Calculated

Each parameter is primarily calculated based on the following three factors:

1. Initial values
2. **Pre-battle buffs**
3. In-battle buffs

How Damage Bonuses Are Calculated

Category	Symbol	Breakdown
Initial values	B_Y	Common value: 100
Pre-battle buffs	a_Y	Disc 5 (Main), Disc 2 Set Effects, etc.
In-battle buffs	b_Y	Disc 4 set effects, ally buffs, Deadly Assault buffs, etc.

$$\text{Damage Bonus } Y = B_Y + a_Y + b_Y$$

Ye Shunguang (Dialyn, Zhao) Example

Category	Value	Breakdown
Initial values	100	Common value : 100
Pre-battle buffs	10	White Water Ballad, 2 sets : 10
In-battle buffs	146	Ye Shunguang Sign : 25 Ye Shunguang Core Passive : 25 Dialyn Core Passive : 40 Zhao Additional Ability : 40 Astral Voice 4 piece (Double) : 16

$$\text{Damage Bonus } Y = 100 + 10 + 146 = 256$$

How Critical Damage Is Calculated

Category	Symbol	Breakdown
Initial values	B_Z	Common Value 150, Core Skill
Pre-battle buffs	a_Z	Disk 4 Main, Disk 2 Set Effect, Sub-Status, etc.
In-battle buffs	b_Z	Disc 4 set effects, ally buffs, Deadly Assault buffs, etc.

$$\text{Critical Damage } Z = B_Z + a_Z + b_Z$$

Ye Shunguang (Dialyn, Zhao) Example

Category	Value	Breakdown
Initial values	150	Common value : 150
Pre-battle buffs	115.2	Ye Shunguang Sign: 48 Substat : 67.2
In-battle buffs	115	Ye Shunguang Sign : 25 Dialyn Sign : 30 Zhao Sign : 30 KoS 4 sets : 30

$$\text{Critical Damage } Z = 150 + 115.2 + 115 = 380.2$$

Attack Power

Category	Symbol	Target
Initial values	B_x	Base Attack + Core Skill + Weapon Attack for each character
Pre-battle Multiplier Buff	a_x	Main effects of Discs 4- 6, Disc 2 set effects, sub-statuses, etc.
Battle Multiplier Buff	b_x	Disc 4 Set Effect, ally buffs (*1), Deadly Assault buffs, etc.
Pre-battle Buff	c_x	Sub-stat and actual values, etc.
Battle Buff	d_x	Allied buffs (*2), etc.

$$\text{ATK } X = \text{Floor}(B_x a_x + c_x) b_x + d_x$$

*1 The crowd accident involving the aggressive Bombardam, the 4-set effect of Woodpecker Electro, etc.

*2 Astra Yao “Graceful Andante”, Yuzuha “Whimsical Wonder” etc.

Ye Shunguang (Dialyn, Zhao) Example

Category	Value	Breakdown
Base Attack Power	1681	Base Attack Power : 863 Core Skill 75 Ye Shunguang Sign: 743
Pre-battle Multiplier Buff	1.97	Substat: 27 Astral Voice 2 sets : 10 Disk 5 and 6 (Main): 60
Battle Multiplier Buff	1.2	Zhao Sign: 10 White Water Ballad 4 sets: 10
Pre-battle Buff	335	Disk 2 Main: 316 Sub-stat actual value: 19
Battle Buff	1000	Zhao Core Passive : 1000

$$\text{ATK X} = \text{Floor}(1681 \times 1.97 + 335) \times 1.2 + 1000 = 5375.2$$

Conditions for Increased Damage

Determine the conditions for increased damage. I mentioned that you could just aim for balance, but it isn't realistic to keep balancing it this way.

$$(X,Y,Z) = (5375.2,256,380.2) \rightarrow (2003.8,2003.8,2003.8)???$$

For each substage 1 hit, the weight differs for each parameter.



Balance the values by eliminating this difference in weight

Comparison of Damage Bonus and Critical Damage

Item	Sub-stage 1 hit	Two-set effect	Disk-based
Damage Bonus	-	10%	30%
Critical Damage	4.8%	16%	48%

→The ratio is the same when using the 2-set effect and the disc as the main weapon

$$\text{Damage Bonus} : \text{Critical Damage} = 10 : 16 = 30 : 48 = 3 : 4.8$$

Damage Bonus and Critical Damage Weighting

Damage Bonus weight: $W_Y = 3$

Critical Damage weight: $W_Z = 4.8$

As mentioned earlier, because the ratios match, the two-set damage bonus applies to items that provide a 10% increase, such as the Fanged Metal series.

The Weight of ATK

Let the initial ATK be B_x and the combat multiplier buff be b_x .

ATK Weight: $W_x = B_x \times 0.03 \times b_x$

Example

In the situation described earlier, the weight of ATK is

$$1681 \times 0.03 \times 1.2 = 60.516.$$

Each item (value/weight)

Item	Value	Weight	Weighted value
ATK	5375.2	60.516	88.822...
Damage Bonus	256	3	85.333...
Critical Damage	380.2	4.8	79.208...

Just switch from ATK% to Critical Damage!

- Sub-stat 1 hit: Migration amount 1 (?)
- 2-set effect: Migration amount $\frac{10}{3}$ (?)
- Main stats 4 and 5: Migration amount 10 (?)

When ATK is involved, the scaling doesn't work out quite right. Try changing one of the sub-stats from ATK% to Critical Damage, and you'll see

Item	Value	Weighted value	Before the migration
ATK	5315.2	87.831...	88.822...
Damage Bonus	256	85.333...	85.333...
Critical Damage	385.2	80.208...	79.208...

Consequently, while critical damage has increased by 1, the decrease in the value divided by the ATK weight is less than 1. **In other words, the sum of the three factors is not maintained before and after the change!**

As a result, we cannot perform the operation of transferring values from larger to smaller numbers, as described in the mathematical background.

The sum is not preserved because decimal places are rounded during the calculation of ATK.

However, since damage is actually increasing, this serves as a sufficient indicator even at this stage.

We will mathematically derive the exact conditions for increasing damage while resolving this issue.

Simulated ATK

Category	Symbol	Target
Default value	B_x	Base ATK by Character + Core Skill + Weapon ATK
Pre-battle Multiplier Buff	a_x	Discs 4–6 (Main) Two-Disc Set Bonus Sub-stat, etc.
Battle Multiplier Buff	b_x	The Four-Disc Set Effect Allied buff Deadly Assault Buff, etc.
Pre-battle buff	c_x	Sub-stat and actual values, etc.
Battle Buffs	d_x	Allied buffs, etc.

$$\text{ATK X} = \text{Floor}(B_x a_x + c_x) b_x + d_x$$

$$\text{Simulated ATK X} = (B_x a_x + c_x) b_x + d_x$$

Using simulated ATK allows you to maintain the transfer amount

Each item (value/weight)

Item	Value	Weight	Weighted value
Simulated ATK	5375.884	60.516	88.834...
Damage Bonus	256	3	85.333...
Critical Damage	380.2	4.8	79.208...

If you replace one sub-stat from ATK% with Critical Damage

Item	Value	Weight	Weighted value
Simulated ATK	5315.368	60.516	87.834...
Damage Bonus	256	3	85.333...
Critical Damage	385	4.8	80.208...

The conversion values for Attack, Damage Bonus, and Critical Damage into hit counts are defined below.

Equivalent number of sub-hits

$$\text{ATK} : h_x = \frac{\tilde{x}}{w_x}$$

$$\text{Damage Bonus} : h_y = \frac{y}{w_y}$$

$$\text{Critical Damage} : h_z = \frac{z}{w_z}$$

- Sub-stage 1 hit: Migration amount 1
- Two-set effect: Migration amount $\frac{10}{3}$
- No. 4 and No. 5 Main: Migration amount 10

Basic Damage (displayed as a value based on the number of sub-hits)

$$D(h_x, h_y, h_z) = C'(\text{Floor}(\frac{w_x h_x - d_x}{b_x}) b_x + d_x) h_y h_z$$

Balance the stats based on the sub-hit count conversion value, and investigate the conditions for increasing Basic damage.

Δ -Migration

The operation of shifting by δ from the i -th component to the j -th component of vector $x = (x_1, x_2, x_3)$ is referred to as the Δ -Migration from i to j , denoted as $\Delta_{i \rightarrow j}(x, \delta)$.

However, assume that $x_i - \delta > x_j + \delta$ is always satisfied.

A set $x = (h_x, h_y, h_z)$ of converted values of substate hit counts for the Δ -Migration $y = \Delta_{i \rightarrow j}(x, \delta)$, $D(y) > D(x)$ Determine the conditions under which this occurs. There are three possible cases to consider:

- Migration between Damage Bonus and Critical Damage ($\{i, j\} = \{2, 3\}$)
- Migration between ATK and Damage Bonus ($\{i, j\} = \{1, 2\}$)
- Migration between ATK and Critical Damage ($\{i, j\} = \{1, 3\}$)

Migration between Damage Bonus and Critical Damage ($\{i, j\} = \{2, 3\}$)

There are two possible scenarios:

When migrating from Damage Bonus to Critical Damage, $(i, j) = (1, 2)$, and when migrating from Critical Damage to Damage Bonus, $(i, j) = (2, 1)$. In both cases, as explained in the section on the mathematical background, $D(y) > D(x)$ holds for any δ .

Note that although I wrote “any δ ,” the condition that the order relations remain unchanged is necessary.

Combining these two conditions,

$$|h_y - h_z| > 2\delta$$

It becomes.

Migration between ATK and Damage Bonus ($\{i, j\} = \{1, 2\}$)

A sufficient condition for the difference $D(h_x - \delta, h_y + \delta, h_z) - D(h_x, h_y, h_z)$ to be greater than or equal to 0 is

$$0.03 B_x \delta (h_x - h_y - \delta) - (h_y + \delta) > 0 \quad (1)$$

So, if we reverse the sign of δ ,

$$0.03 B_x \delta (h_y - h_x - \delta) - (h_y + \delta) > 0 \quad (2)$$

Therefore, when combining these two, the sufficient condition is to treat $\pm\delta$ as $+\delta$.

$$0.03 B_x \delta (|h_y - h_x| - \delta) - (h_y + \delta) > 0 \quad (3)$$

Minimum value of the initial ATK

The minimum base ATK at level 60 is 568 for Seth. Additionally, the weapon ATK at level 60 must be at least 594, the minimum value for an A-rank W-Engine, and when you add 25 each from Core Skill at B, D, and F,

$$B_x \geq 568 + 75 + 594 = 1237.$$

Applying this to Equation (3) and simplifying, we get

$$\delta(0.03 \times 1237(|h_x - h_y| - \delta) - 1) > h_y \quad (4)$$

Obtain it. Evaluate this expression according to the following two approaches.

1. It is good if the migration amount δ is large.
2. Even if the migration amount δ is small, it is sufficient as long as the difference $|h_x - h_y|$ is large.

Migration amount δ

The migration amount shall be either $\delta = 1$ or $\delta \geq \frac{10}{3}$.

When the migration amount is $\delta = 1$

Equation (4) takes the following form:

$$0.03 \times 1237(|h_x - h_y| - 1) - 1 > h_y. \quad (5)$$

$ h_x - h_y $	$0.03 \times 1237(h_x - h_y - 1) - 1$
2	36.11
3	73.22
4	110.33
5	147.44

The rest is a matter of calculation: setting the difference $|h_x - h_y| \geq 5$, we have $147.44 > h_y$. The damage bonus corresponding to this is 442.32.

When the migration amount is $\delta \geq \frac{10}{3}$

Since the condition for maintaining the order relationship implies that $|h_x - h_y| - \delta > \delta$, adding this term and rearranging equation (3) yields

$$\delta(0.03 \times 1237\delta - 1) > h_y. \quad (5)$$

That would be good.

Since we set the migration amount δ to $\frac{10}{3}$ or greater, the left-hand side must be 409 or greater. Therefore, the condition we obtain is

$$409 > h_y.$$

This is weaker than the condition when the migration amount is 1.

Migration between ATK and Critical Damage ($\{i,j\}=\{1,3\}$)

Simply replace the hypothetical h_y from earlier with h_z .

A sufficient condition for the difference $D(h_x \pm \delta, h_y, h_z \mp \delta) - D(h_x, h_y, h_z)$ (with compatible order) to be nonnegative is

$$0.03B_x\delta(|h_x - h_z| - \delta) - (h_z - \delta) > 0.$$

When the migration amount is $\delta = 1$

Let the difference $|h_x - h_z|$ be ≥ 5 ,

$$147.44 > h_z.$$

When the migration amount is $\delta \geq \frac{10}{3}$

$$409 > h_z.$$

Conditions for Increased Basic Damage

Set the base ATK to 1237 or higher, the damage bonus to 442 or lower, and the critical damage to 707 or lower. At this point, the conversion values for the number of sub-hits related to basic damage, $x = (h_x, h_y, h_z)$, and let Δ -Migration $y = \Delta_{i \rightarrow j}(x, \delta)$ satisfy one of the following conditions.

- $\{i, j\} = \{1, 2\}$, $\delta = 1$, and $|h_x - h_y| \geq 5$
- $\{i, j\} = \{1, 3\}$, $\delta = 1$, and $|h_x - h_z| \geq 5$
- $\{i, j\} = \{1, 2\}$, and $\delta \geq \frac{10}{3}$
- $\{i, j\} = \{1, 3\}$, and $\delta \geq \frac{10}{3}$
- $\{i, j\} = \{2, 3\}$

In this case, $D(y) > D(x)$.

Example of Ye Shunguang

Item	Number of sub-hits	Two-set effect	No. 4 and 5 Disk (Main)
ATK	9	Astral Voice	No. 5
Damage Bonus	-	White Water Ballad	-
Critical Damage	14	-	-

Item	Value	Weight	Equivalent hits
Simulated ATK	5375.884	60.516	88.834...
Damage Bonus	256	3	85.333...
Critical Damage	380.2	4.8	79.208...

Example of Basic Damage Calculation

Item	Value	Notes
ATK	5375.2	Please refer to the previous slides for the breakdown
Damage Bonus	2.56	Please refer to the previous slides for the breakdown
Critical Damage	3.802	Please refer to the previous slides for the breakdown
Defense Adjustment	$\frac{794}{1706.04}$	VR Training Center Dullahan, Level 70, Flat PEN 9
Damage Multiplier	38.5	Ultimate: Chasing Storms, Level 12
Damage Resistance	1.2	Initial value: 100 Ye Shunguang Sign: 20
Stun DMG Multiplier	1.8	Initial value: 150 Dialyn Core Passive: 30 Veil Vulnerability Applied
Damage Calculation	2024848.243...	The result of multiplying all the values of the items above
Displayed Damage	2024849	Values displayed in the game

Replace 2 sets of 【Astral Voice】 with 【Branch & Blade Song】

The difference between the ATK and Critical Damage values, when converted to hit points, is 9 or more.

The difference between the ATK and Critical Damage values, when converted to hit counts, is 9 or more. If you change 【Astral Voice】 to 【Branch & Blade Song】 , $\frac{10}{3}$ of your ATK will be converted to Critical Damage.

Item	Value	Weight	Equivalent hits
Simulated ATK	5174.164	60.516	85.500...
Damage Bonus	256	3	85.333...
Critical Damage	396.2	4.8	82.541...

The calculation damage just now is 2030921.13..., and you can see that the damage is increasing.

In Conclusion

- Finally, this is a claim based on a mathematical sufficient condition; there are many cases where damage increases simply by maintaining balance, even if the condition is not met.
- Although the minimum value was adopted for the base ATK, the conditions become a bit weaker when targeting attackers with higher base ATK.
- While there are exceptions—such as Soldier 0 - Anby, whose additional attacks use a different method for calculating critical damage—these do not currently hinder the discussion.
- The condition of maintaining the number of substat hits is realistically difficult, so please treat the 2-set effect and the indicators for selecting the 4th and 5th main disk options as a reference only.
- The mechanics for Anomaly Damage and Disorder Damage are nearly identical, but Sheer Force Damage works slightly differently because it focuses on HP.